

3(a)	1 both alleles of a, gene / genotype, are the same ; 2 phenotype / effect, of (recessive) allele is masked by a dominant allele or two copies of the (recessive) allele are needed for phenotype to be displayed or alleles only affect the phenotype in the absence of a dominant allele or (genotype) must be homozygous for the (recessive) allele to affect the phenotype ;	2
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Question	Answer	Marks
3(b)(i)	number of heterozygous people = 1 680 000 ; ; ; must be to 3 significant figures or more A 1 675 000 to 1 685 000 ; ; ; R 1 700 000 max 2 if decimal places in final answer allow max 2 for working if final answer incorrect ($q = \sqrt{10\,800 / 67\,100\,000}$ or $\sqrt{0.000161}$ or 0.0127) $p = 1 - 0.0127$ or 0.987 ($2pq = 2 \times 0.9873 \times 0.0127$ or 0.0251) calculation for number of heterozygous people = $67\,100\,000 \times 0.0251$ or $2pq$	3
3(b)(ii)	any three from: 1 ref. natural selection ; 2 selection pressure against cystic fibrosis (CF) ; 3 (people with) CF do not survive as long / AW ; 4 heterozygotes are calculated from a smaller CF population / AW ; 5 (some) people with CF who have died will have passed on CF allele ;	3

Question	Answer	Marks
3(c)(i)	any three from: 1 (median) predicted life expectancy increased (over time) ; 2 data quote ; e.g. from 2008 to 2020 increase from 38.8 to 50.6 years increase of 11.8 % increase of 30.4 3 treatment can be started early ; 4 better treatment / improved care (between 2008 and 2020) ;	3
3(c)(ii)	any three from: 1 reduces worry / AW, if result is negative ; ora 2 can make informed reproductive decisions / AW ; 3 cost / availability, of test ; 4 plan for care of child with CF ; 5 <i>idea of</i> further genetic testing ; e.g. counselling / testing of embryo / testing of partners 6 <i>idea that</i> test is not 100% accurate ; 7 problems related to, stigma / discrimination / insurance / confidentiality ;	3